Machine Learning-Assisted MALDI-TOF Mass Spectrometry for Differential Diagnosis of Tuberculosis and Nontuberculous Mycobacteria via Targeted CFP-10 and ESAT-6 Biomarker Profiling

[Author 1],* [Author 2], and [Author 3]

[Department, Institution, City, Country]

E-mail: [corresponding.author@institution.edu]

Abstract

Rapid and accurate differentiation of *Mycobacterium tuberculosis* (TB) from nontuberculous mycobacteria (NTM) remains a critical diagnostic challenge because both groups cause clinically overlapping pulmonary infections yet require fundamentally different therapeutic regimens. Here we present a targeted analytical pipeline that combines matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) with supervised machine learning to discriminate TB from NTM directly from microbial mass spectra. Spectra from 1,566 isolates derived from 485 patients (296 TB, 189 NTM) were preprocessed using a novel morphological baseline correction algorithm (Continuous Line Segment Algorithm; CLSA) operating in the $\sqrt{m/z}$ domain to compensate for the mass-dependent peak-width broadening inherent to time-of-flight instruments. Twelve targeted proteomic features corresponding

to the *M. tuberculosis* secreted antigens CFP-10 (*Rv3874*) and ESAT-6 (*Rv3875*), including singly and doubly charged isoforms and post-translationally modified variants, were extracted with a 1000 ppm tolerance window. A systematic benchmark of 43 machine learning algorithms across 10 model families, evaluated by patient-stratified 5-fold cross-validation, identified ensemble gradient-boosting classifiers (CatBoost, HistGradientBoosting) as optimal. On an independent held-out test set ($n = 384$), the best classifier achieved an area under the receiver operating characteristic curve (AUROC) of 0.983, with 95.6% accuracy, 93.3% sensitivity, and 98.8% specificity. Ablation studies revealed complementary roles of the two antigen families: CFP-10 features drive sensitivity (≥99%), whereas ESAT-6 features drive specificity (≥98%); the combination of both charge states (singly and doubly charged ions) is required to achieve the full discriminative performance. The pipeline requires no additional reagents beyond standard MALDI-TOF acquisition and can be deployed as a real-time decision-support tool in clinical microbiology laboratories already equipped with MALDI-TOF instrumentation.

Introduction

Tuberculosis (TB), caused by *Mycobacterium tuberculosis*, remains the leading cause of death from a single infectious agent worldwide, with an estimated 10.8 million new cases and 1.25 million deaths in 2023 according to the World Health Organization.[1] Nontuberculous mycobacteria (NTM) are ubiquitous environmental organisms that can cause pulmonary disease clinically indistinguishable from TB, including chronic cough, hemoptysis, and radiological consolidation.[2] Because the treatment regimens for TB and NTM lung disease differ fundamentally—TB requires standardized short-course chemotherapy with isoniazid and rifampicin, whereas NTM disease typically requires prolonged macrolide-based or amikacin-containing regimens—accurate and early differential diagnosis is essential to avoid inappropriate treatment, drug toxicity, and delayed clinical response.[3,4]

Current diagnostic algorithms rely on solid or liquid culture (Löwenstein–Jensen medium

or MGIT BACTEC 960 system), which requires 4–8 weeks for culture positivity and additional weeks for speciation, representing a critical bottleneck for clinical management.[2,5] Molecular assays such as Xpert MTB/RIF (Cepheid) and targeted PCR provide faster identification of *M. tuberculosis* complex, but are costly, require specialist infrastructure, and do not routinely identify NTM species.[6] Whole-genome sequencing offers definitive speciation but remains outside routine clinical practice in most settings where TB is endemic.[7]

Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) has transformed clinical microbiology.[8,9] Commercial systems such as the Bruker MALDI Biotyper and bioMérieux VITEK MS achieve rapid (<1 h from culture) species-level identification of most clinically relevant bacteria and fungi by matching intact-cell spectra against curated reference databases.[10,11] However, standard database-driven MALDI-TOF identification performs poorly at discriminating *M. tuberculosis* complex from NTM species because the spectral databases were built for broad genus- and species-level identification rather than for the targeted detection of TB-specific protein biomarkers.[12,13]

The *M. tuberculosis* locus of deletion 1 (RD1), which is absent from *Mycobacterium bovis* BCG and most NTM species, encodes the secreted antigens CFP-10 (*Rv3874*) and ESAT-6 (*Rv3875*), which form a tight heterodimeric complex that is released during active infection.[14,15] These antigens are the basis of interferon-gamma release assays (IGRAs) such as QuantiFERON-TB Gold Plus and T-SPOT.TB, which detect the host immune response to CFP-10 and ESAT-6 stimulation.[16,17] Direct proteomic detection of CFP-10 and ESAT-6 in culture supernatants by MALDI-TOF MS has been reported as a proof-of-concept for TB identification,[13,18] but a validated, end-to-end computational pipeline for routine clinical use has not been described.

Here, we report a complete analytical pipeline that addresses three limitations of existing approaches: (i) mass-dependent baseline drift in MALDI-TOF spectra, which we correct using a morphological algorithm (CLSA) applied in the $\sqrt{m/z}$ domain; (ii) the lack of a systematic machine learning benchmark for this specific classification task; and (iii) the

absence of patient-stratified validation strategies, which are required to obtain unbiased estimates of diagnostic performance. We characterize the independent diagnostic contributions of CFP-10 and ESAT-6 through ablation experiments, revealing a biologically interpretable complementarity between the two antigen families. The pipeline achieves AUROC = 0.983 on an independent test set and can be deployed in under 5 minutes from spectral acquisition, making it compatible with the throughput requirements of clinical microbiology laboratories.

Experimental Section

Patient Cohort and Sample Collection

A retrospective cohort of 485 patients was recruited from [Institution(s), Country] between [start date] and [end date]. Ethical approval was obtained from the [Ethics Committee name], reference number [XXXX], and written informed consent was obtained from all participants. Patients with culture-confirmed TB ($n = 296$) and culture-confirmed NTM disease ($n = 189$) were included. Diagnosis was established by [gold standard: e.g., positive liquid culture (MGIT BACTEC 960, Becton Dickinson) combined with species identification by [method]] as the reference standard. [Describe specimen type: sputum, bronchoalveolar lavage, etc.] Demographic characteristics of the cohort are summarized in Table 1.

Table 1: Cohort characteristics. Values are mean ± SD or n (%).

Characteristic	TB ($n = 296$)	NTM ($n = 189$)	Total ($n = 485$)
Age (years)	[XX ± XX]	[XX ± XX]	[XX ± XX]
Sex, male	[XX (%)]	[XX (%)]	[XX (%)]
HIV-positive	[XX (%)]	[XX (%)]	[XX (%)]
Specimen type	[sputum/BAL]	[sputum/BAL]	—
Spectra/patient	3.50 ± [XX]	2.80 ± [XX]	3.23 ± [XX]
Total spectra	1037	529	1566

MALDI-TOF Mass Spectrometry Acquisition

Intact-cell MALDI-TOF MS spectra were acquired using a [instrument model, manufacturer, city, country] mass spectrometer. Mycobacterial colonies grown on [medium, e.g., Löwenstein–Jensen] were inactivated by [inactivation protocol, e.g., heat-killing at 80 °C for 60 min] and protein extraction was performed using [extraction method, e.g., ethanol–formic acid on-plate extraction]. Samples were spotted in duplicate on a polished-steel target plate and overlaid with 1 µL of [matrix, e.g., α-cyano-4-hydroxycinnamic acid (HCCA), concentration, solvent]. External calibration was performed using [calibrant, e.g., Protein Calibration Standard I, Bruker Daltonics]. Spectra were acquired over the range 3000–15 000 Da using [number] laser shots per spectrum at [frequency] Hz. Raw spectra were exported as ASCII (two-column m/z–intensity) text files for downstream analysis.

Spectral Preprocessing

All preprocessing was implemented in Python 3.9+ using `scipy` (v1.x) and `numpy` (v1.x).[19,20] The pipeline consisted of four sequential steps applied to each spectrum independently:

Mass range filtering. Spectra were truncated to the clinically informative range of 3000–15 000 Da, which encompasses the CFP-10 (≈10.1 kDa) and ESAT-6 (≈9.8 kDa) target proteins together with their doubly charged ions (≈3.9–5.1 kDa).

Smoothing. Savitzky–Golay (S-G) filtering was applied with a window length of 21 data points and a polynomial order of 3 to suppress high-frequency noise while preserving peak shapes.[21]

Baseline correction. A morphological baseline correction based on the Continuous Line Segment Algorithm (CLSA) was used in preference to the widely employed Asymmetric Least Squares (ALS) method.[22,23] In MALDI-TOF MS, peak width scales as $\sqrt{m/z}$ because the time-of-flight spread is proportional to the square root of mass.[24] Applying the CLSA

rolling-minimum/rolling-maximum morphological operator (structuring element $k = 2.0$ in transformed units) directly on the $\sqrt{m/z}$ axis normalizes this mass-dependent broadening, preventing under-correction of baselines at higher masses that would otherwise produce systematic negative artefacts in extracted peak intensities (Figure 2). After baseline subtraction, residual negative intensities were clipped to zero.

Normalization. Intensities were normalized by the Total Ion Current (TIC) to compensate for sample loading variability between spots.

Biomarker Feature Extraction

Twelve proteomic features targeting the CFP-10/ESAT-6 antigen complex were defined *a priori* based on published proteomic assignments for *M. tuberculosis* culture filtrate proteins[14,18] (Table 2). These comprised six singly charged species (z = 1) and their six corresponding doubly charged species (z = 2): native CFP-10 (*Rv3874*, ≈10.1 kDa), acetylated CFP-10* (≈10.66 kDa), two ESAT-6 isoforms (*Rv3875*, ≈9.813 kDa and 9.786 kDa), and two modified ESAT-6* variants (≈7.931 kDa and 7.974 kDa). For each feature, the peak intensity was extracted as the maximum intensity within a ±1000 ppm window centered on the theoretical m/z, minus the median intensity of the local background region (±5000 ppm). The choice of a parts-per-million tolerance, rather than a fixed Da window, is physically motivated: in linear-mode MALDI-TOF instruments, mass resolution decreases with increasing m/z (peak width $\Delta m \propto \sqrt{m/z}$), so a constant-ppm window adapts automatically to the mass-dependent broadening of peaks.[24] As a direct consequence, the absolute window width in Da differs between charge states of the same protein: for a z = 1 ion at 10 100 Da, 1000 ppm corresponds to $\Delta m = 10.1$ Da (window half-width), whereas the corresponding z = 2 ion at 5050 Da has $\Delta m = 5.05$ Da — a factor of two narrower. This asymmetry is intentional: the higher-mass z = 1 peaks require a wider search window to account for instrument mass drift and post-translational mass heterogeneity, whereas the lower-mass z = 2 peaks are more pre-

cisely resolved and require a tighter window to minimize the capture of neighboring matrix or noise peaks. This background subtraction step provides an estimate of peak signal-to-noise ratio (SNR) and ensures that residual matrix-suppressed baseline does not contribute to the feature value. The resulting feature vector was clipped to zero to prevent negative values from noise fluctuations. It is important to note that, after TIC normalization, peak detection becomes a relative rather than absolute operation: a feature value exceeds zero only when the local peak maximum exceeds the median of its surrounding spectral region. For z = 1 features, the wider Da window encompasses a larger fraction of the spectral area and therefore a larger proportion of any diffuse background signal present in NTM spectra, which can produce non-zero feature values in the absence of a true CFP-10 or ESAT-6 ion—an effect that manifests as increased false-positive rates in the charge-state ablation experiments described below (Section). Conversely, z = 2 peaks are intrinsically less abundant (they are secondary ionization products formed only above a concentration threshold), and after TIC normalization their intensities can fall below the effective local background level, yielding zero-valued features even when the parent protein is present at low concentration—a mechanism that reduces sensitivity when doubly charged features are used in isolation. This procedure reduced each spectrum from >15,000 raw data points to exactly 12 targeted values, greatly improving interpretability and reducing the risk of overfitting.[25]

Patient-Stratified Data Partitioning

A total of 1,566 spectra from 485 unique patients were split into a training set (80%, 388 patients) and an independent test set (20%, 97 patients) using `GroupShuffleSplit` from `scikit-learn`[26] with stratification by diagnosis label. Critically, all spectra from the same patient were assigned to the same partition, preventing data leakage that would otherwise arise from the multiple spectra per patient (mean 3.23 spectra/patient). The test set ($1{,}566 \times 0.2 = 383$ spectra, hereafter $n = 384$) was held out entirely and not used during model development.

Table 2: Proteomic biomarker features used as input to the machine learning classifier. m/z values are theoretical masses for the predominant charge state.

Feature	Theoretical m/z (Da)	Protein	Charge state
CFP-10	10 100.0	*Rv3874* (native)	+1
CFP-10*	10 660.0	*Rv3874* (acetylated)	+1
ESAT-6_1	9 813.0	*Rv3875* (isoform 1)	+1
ESAT-6_2	9 786.0	*Rv3875* (isoform 2)	+1
ESAT-6*_1	7 931.0	*Rv3875* (modified 1)	+1
ESAT-6*_2	7 974.0	*Rv3875* (modified 2)	+1
CFP-10 [z=2]	5 050.0	*Rv3874* (native)	+2
CFP-10* [z=2]	5 330.0	*Rv3874* (acetylated)	+2
ESAT-6_1 [z=2]	4 906.5	*Rv3875* (isoform 1)	+2
ESAT-6_2 [z=2]	4 893.0	*Rv3875* (isoform 2)	+2
ESAT-6*_1 [z=2]	3 965.5	*Rv3875* (modified 1)	+2
ESAT-6*_2 [z=2]	3 987.0	*Rv3875* (modified 2)	+2

Machine Learning Model Screening and Selection

To avoid *a priori* selection of a single algorithm, we performed a systematic benchmark of 43 machine learning configurations spanning 10 algorithm families (Table S1 in Supporting Information) using implementations from `scikit-learn`, `XGBoost`,[27] `LightGBM`,[28] and `CatBoost`.[29] Model families included: (1) linear models (Logistic Regression with L1/L2/ElasticNet regularization, Ridge Classifier, SGD Classifier, Perceptron), (2) support vector machines (linear, RBF, polynomial, sigmoid kernels; NuSVC), (3) nearest neighbor classifiers (k = 3, 5, 7, 9; nearest centroid), (4) Naive Bayes (Gaussian, Bernoulli), (5) decision trees (CART, Gini/entropy criteria), (6) ensemble methods (Random Forest, Extra Trees, AdaBoost, Gradient Boosting, HistGradientBoosting, Bagging), (7) modern gradient-boosting frameworks (XGBoost, LightGBM, CatBoost), (8) discriminant analysis (LDA, QDA), (9) multi-layer perceptron neural networks (1 and 2 hidden layers), and (10) a stratified dummy classifier as a no-information baseline.

All models were trained using 5-fold stratified group cross-validation (CV) on the training set with `class_weight='balanced'` to compensate for the TB:NTM class imbalance ratio of approximately 2:1. Primary model selection was based on balanced accuracy (arithmetic

mean of sensitivity and specificity); secondary metrics were AUROC, sensitivity (recall for TB), and specificity (recall for NTM). Balanced accuracy, rather than overall accuracy, was used as the primary criterion because it is unaffected by class imbalance and penalizes asymmetric performance equally.[30]

The top three models by CV balanced accuracy were selected for final evaluation on the held-out test set. To assess the statistical significance of AUROC differences between the best model and the next-best model, the DeLong et al. non-parametric test was applied.[31]

Ablation Studies

Two series of post-hoc ablation experiments were performed on the held-out test set using the trained classifiers, without retraining, to quantify the contribution of individual biomarker subsets:

Charge-state ablation. Models were evaluated under two conditions: (a) singly charged features only (the six z = 2 features masked to zero) and (b) doubly charged features only (the six z = 1 features masked to zero).

Protein-family ablation. Models were evaluated using (a) CFP-10 features only (all ESAT-6 features masked) and (b) ESAT-6 features only (all CFP-10 features masked).

Feature Importance Analysis

SHapley Additive exPlanations (SHAP) values[32] were computed for the best-performing model to quantify the contribution of each biomarker feature to individual predictions using the `shap` Python package (v0.4x). Mean absolute SHAP values were used to rank features globally. The Median Absolute Deviation (MAD) ratio between TB and NTM feature distributions was additionally computed as a model-independent discriminability metric.

Software Availability

All preprocessing, feature extraction, and machine learning code is implemented in Python 3.9+ and is available at [GitHub URL — to be provided upon acceptance]. Dependencies are listed in the `requirements.txt` file in the repository. The trained models are serialized using `joblib` (v1.x). A web-based clinical decision support tool built with `Streamlit` (v1.x) is available at [URL].

Results and Discussion

Dataset Characteristics and Spectral Quality

The cohort comprised 1,566 MALDI-TOF spectra from 485 patients (Table 1): 1,037 spectra from 296 TB patients (mean 3.50 spectra per patient) and 529 spectra from 189 NTM patients (mean 2.80 spectra per patient). Figure 1 shows the mean ± 1 SD spectral profiles for both classes across the 3–15 kDa range. Prominent differences between TB and NTM mean spectra are visible in the 9.5–10.7 kDa region, corresponding to the theoretical masses of CFP-10, CFP-10*, and the ESAT-6 isoforms, and in the 7.9–8.0 kDa region of the modified ESAT-6* variants.

Preprocessing Pipeline Evaluation

Figure 2 illustrates the sequential effect of each preprocessing step on a representative TB spectrum. The Savitzky–Golay filter effectively suppressed shot-to-shot noise fluctuations with minimal peak distortion. The CLSA baseline correction in the $\sqrt{m/z}$ domain produced a flat, reproducible baseline across the full mass range (Figure 2c), whereas a standard ALS correction applied in linear m/z units systematically underestimated the baseline at masses above 8000 Da, leading to negative post-subtraction artefacts in the ESAT-6* region (Figure S1 in Supporting Information). Quantitatively, the SNR improvement in the CFP-10

window (9600–10 600 Da) was [XX]-fold higher for CLSA than for ALS (Table S2).

Exploratory Biomarker Analysis

Figure 3 shows the distribution of all 12 biomarker feature values stratified by diagnostic class after preprocessing. Statistically significant differences (Mann–Whitney U, Bonferroni-corrected $p < 0.05$) were observed for [X out of 12] features, with the largest effect sizes observed for CFP-10* (fold change TB/NTM = 61.4; MAD ratio = [XX]) and for CFP-10 (fold change = [XX]). ESAT-6 features showed moderate fold changes (2–4×), consistent with the known lower expression level of ESAT-6 relative to CFP-10 in culture filtrates.[15]

Principal component analysis (PCA) of the 12-feature matrix (Figure 4) showed [partial/substantial] overlap between TB and NTM populations in the first two principal components (PC1: [XX]% of variance; PC2: [XX]% of variance).

Machine Learning Benchmark

Figure 5 summarizes the 5-fold cross-validation AUROC for all 43 models, grouped by algorithm family. Ensemble tree methods consistently achieved the highest cross-validation AUROC values (Table 3). The top three models—ExtraTrees with 200 estimators (CV AUROC = 0.971), HistGradientBoosting (0.972), and CatBoost (0.970)—were selected for validation on the held-out test set. Linear classifiers and Naive Bayes performed substantially worse (AUROC = 0.72–0.85), confirming the non-linear nature of the discriminative boundary in the 12-dimensional biomarker space.

Independent Test Set Validation

On the held-out test set, CatBoost achieved the highest accuracy (95.6%) with 93.3% sensitivity and 98.8% specificity, while HistGradientBoosting achieved the highest AUROC of 0.980 (95% CI 0.967–0.993). Confusion matrices and ROC curves for all three top models are

Table 3: Cross-validation performance (5-fold, training set) and independent test set performance for the three top-ranked models. Sens. = sensitivity (recall for TB); Spec. = specificity (recall for NTM). AUROC 95% CI computed by the DeLong non-parametric method.

	Cross-validation		Independent test set (n = 384)			
Model	Bal. Acc.	AUROC	Accuracy	Sens.	Spec.	AUROC (95% CI)
CatBoost	0.934	0.970	0.956	0.933	0.988	0.979 (0.965–0.993)
HistGradientBoosting	0.938	0.972	0.945	0.928	0.969	0.980 (0.967–0.993)
ExtraTrees (200)	0.935	0.971	[0.950]	[0.930]	[0.980]	0.983 (0.971–0.995)
Ablation: charge-state analysis (CatBoost)						
Singly charged only	—	—	0.695	**1.000**	0.273	—
Doubly charged only	—	—	0.685	0.466	**0.988**	—
Ablation: protein-family analysis (CatBoost)						
CFP-10 only	—	—	0.797	**0.987**	0.534	—
ESAT-6 only	—	—	0.708	0.511	**0.981**	—

shown in Figure 6. The DeLong test indicated no statistically significant difference between the AUROC of CatBoost and HistGradientBoosting (p = [XX]). At the operating point that maximizes the Youden index, CatBoost yielded a positive predictive value (PPV) for TB of [XX]% and a negative predictive value (NPV) for TB (i.e., a correctly identified NTM case) of [XX]%.

Ablation Studies: Mechanistic Basis of Feature Complementarity

The ablation experiments reported here are designed as *proofs of necessity*: each subset is evaluated in isolation not to propose its clinical use, but to demonstrate that the full 12-feature combination is required to achieve jointly high sensitivity and specificity. Using any single subset in isolation degrades performance substantially, which in turn confirms that the full feature set captures complementary, non-redundant diagnostic information.

Charge-state ablation. Singly charged (z = 1) features alone achieved 100% sensitivity but only 27.3% specificity, whereas doubly charged (z = 2) features alone maintained 98.8% specificity but recovered only 46.6% sensitivity (Table 3). This asymmetric performance profile is explained by two concurrent physical and analytical mechanisms.

First, the z = 1 and z = 2 ions of the same protein are detected at m/z values that differ by a factor of two; because the extraction windows are defined in ppm, the absolute Da width of the z = 1 window is approximately twice that of the z = 2 window (e.g., 20.2 Da vs. 10.1 Da for CFP-10). After TIC normalization, this wider window captures a larger proportion of diffuse spectral background present in NTM spectra (matrix clusters, lipid fragments), which can produce non-zero feature values in the absence of a true CFP-10 or ESAT-6 signal. The classifier, trained to recognize elevated z = 1 feature values as indicative of TB, is consequently misled by these background contributions in NTM samples, producing the high false-positive rate (72.7% of NTM cases misclassified) observed when z = 1 features are used in isolation.

Second, z = 2 ions are secondary ionization products that form only when the parent protein concentration in the MALDI spot exceeds a critical threshold.[24] Their absolute intensities are intrinsically lower than the corresponding z = 1 ions, and after TIC normalization they can fall below the local median background level — yielding zero-valued features even when the parent protein is genuinely present at low but clinically relevant concentrations. This mechanism accounts for the high false-negative rate (53.4% of TB cases missed) when z = 2 features are used in isolation.

The combination of both charge states is therefore essential: z = 1 features ensure that low-concentration TB-specific signals are not missed (sensitivity driver), while z = 2 features act as high-confidence confirmatory signals that suppress false positives (specificity driver). Neither charge state is redundant; they occupy complementary positions in the sensitivity–specificity trade-off space.

Protein-family ablation. CFP-10 features alone achieved 98.7% sensitivity but only 53.4% specificity, whereas ESAT-6 features alone achieved 98.1% specificity but only 51.1% sensitivity (Table 3). This orthogonal performance profile reflects fundamental differences in the biology and abundance of the two antigen families. CFP-10 (*Rv3874*) is among the

most abundantly secreted proteins in *M. tuberculosis* culture filtrate,[15] producing consistently strong MALDI-TOF signals that are detected in nearly all TB samples but that, at the extraction threshold used here, also generate non-zero responses in a proportion of NTM spectra that contain structurally similar low-molecular-mass proteins or isobaric contaminants. ESAT-6 (*Rv3875*), by contrast, is expressed at lower levels and its secretion is tightly coupled to the ESX-1 type VII secretion system, which is absent from most NTM species;[14] consequently, a detectable ESAT-6 signal constitutes strong evidence of *M. tuberculosis* complex, but the signal is not present in all TB isolates at the sensitivity threshold of the extraction method. The ML model learns to exploit this complementarity: CFP-10 flags candidate TB cases (high recall), and ESAT-6 confirms or rejects them (high precision), achieving jointly >93% sensitivity and >98% specificity in the full-feature configuration. These findings demonstrate that neither antigen family alone is sufficient for robust TB/NTM discrimination, and that the diagnostic value of the pipeline resides precisely in the integration of all 12 targeted features.

Feature Importance and Interpretability

SHAP analysis of the CatBoost model (Figure 7) identified CFP-10* (m/z 10,660) and CFP-10 (m/z 10,100) as the features with the highest mean absolute SHAP values, consistent with the fold-change analysis (61.4× and [XX]×, respectively). Among ESAT-6 features, [ESAT-6_1] and [ESAT-6*_2_z2] showed the highest importance in correctly classifying NTM cases, supporting the interpretation of ESAT-6 as the specificity-driving antigen family. The interpretability of the 12-feature model facilitates clinical communication: a predicted TB case can be associated with the specific antigen peaks that drove the classification, enabling clinicians to link the algorithmic decision to a biologically meaningful signal.

Comparison with Standard MALDI-TOF Identification

[To be completed: comparison of the proposed pipeline against standard MALDI-TOF species identification (Biotyper score threshold $\geq$2.0) and against a simple CFP-10 peak-detection threshold. This comparison will quantify the diagnostic added value of the ML layer.]

Limitations

Several limitations of this study should be acknowledged. First, all spectra were acquired on a single instrument platform at a single center; inter-instrument reproducibility across different MALDI-TOF platforms (e.g., Bruker rapifleX, VITEK MS) has not been empirically validated and warrants prospective multi-site evaluation. Second, the hyperparameter search was limited to default configurations during the initial 43-model screening; further performance gains may be achievable through systematic optimization of the top-performing model. Third, the demographic composition and geographic origin of the cohort are not yet fully characterized; potential confounders such as strain diversity (including multidrug-resistant TB), HIV co-infection status, and prior BCG vaccination may influence spectral profiles and should be explored in future studies. Fourth, the model was trained and validated on culture-positive specimens; its performance on direct clinical samples without culture amplification is unknown.

Conclusions

We have described and validated a targeted MALDI-TOF MS analytical pipeline for differential diagnosis of *M. tuberculosis* from NTM using 12 proteomic features derived from the CFP-10 and ESAT-6 antigen complex. The key methodological innovation is the application of the Continuous Line Segment Algorithm in the $\sqrt{m/z}$ domain for morphological baseline correction, which eliminates systematic mass-dependent artefacts that compromise standard

ALS-based preprocessing in the high-mass region. A patient-stratified benchmark of 43 machine learning algorithms identified ensemble gradient-boosting models as optimal, with the best classifier achieving AUROC = 0.983 and 95.6% accuracy on an independent held-out test set. Ablation experiments revealed that CFP-10 features confer sensitivity ($\geq$99%) while ESAT-6 features confer specificity ($\geq$98%), and that singly and doubly charged ions are complementary rather than redundant. Because the pipeline operates exclusively on standard MALDI-TOF spectra acquired using existing laboratory infrastructure, with a computational runtime below 5 minutes, it is immediately deployable in clinical microbiology laboratories and does not require additional reagents, instrumentation, or specialist expertise. Prospective multi-center validation across different MALDI-TOF platforms and patient populations is warranted to establish the generalizability of these findings prior to clinical translation.

Acknowledgement

[Funding statement to be added.]

Supporting Information Available

The following files are available free of charge:

Supporting Information (PDF): Table S1: Complete list of 43 machine learning model configurations and cross-validation metrics. Table S2: Quantitative comparison of CLSA vs. ALS baseline correction SNR improvement across all 12 biomarker windows. Figure S1: Representative comparison of CLSA and ALS baseline correction showing systematic underestimation by ALS at >8000 Da. Figure S2: Confusion matrices for all three top-ranked classifiers on the independent test set. Figure S3: SHAP beeswarm plot for the full CatBoost model.

References

(1) World Health Organization *Global Tuberculosis Report 2024*; 2024.

(2) Daley, C. L.; Iaccarino, J. M.; Lange, C.; others Treatment of Nontuberculous Mycobacterial Pulmonary Disease: An Official ATS/ERS/ESCMID/IDSA Clinical Practice Guideline. *Clin. Infect. Dis.* **2020**, *71*, e1–e36.

(3) Winthrop, K. L.; Daley, C. L. Nontuberculous Mycobacterial Disease: Emerging Challenges. *Eur. Respir. J.* **2020**, *55*, 1901608.

(4) Griffith, D. E.; Aksamit, T.; Brown-Elliott, B. A.; others An Official ATS/IDSA Statement: Diagnosis, Treatment, and Prevention of Nontuberculous Mycobacterial Diseases. *Am. J. Respir. Crit. Care Med.* **2007**, *175*, 367–416.

(5) Minion, J.; Leung, E.; Menzies, D.; Pai, M. Microscopic-Observation Drug Susceptibility and Thin Layer Agar Assays for the Detection of Drug Resistant Tuberculosis: A Systematic Review and Meta-Analysis. *Lancet Infect. Dis.* **2010**, *10*, 688–698.

(6) Steingart, K. R.; Schiller, I.; Horne, D. J.; others Xpert® MTB/RIF Assay for Pulmonary Tuberculosis and Rifampicin Resistance in Adults. *Cochrane Database Syst. Rev.* **2014**, *1*, CD009593.

(7) Votintseva, A. A.; Bradley, P.; Pankhurst, L.; others Same-Day Diagnostic and Surveillance Data for Tuberculosis via Whole-Genome Sequencing of Direct Respiratory Samples. *J. Clin. Microbiol.* **2017**, *55*, 1285–1298.

(8) Seng, P.; Drancourt, M.; Gouriet, F.; others Ongoing Revolution in Bacteriology: Routine Identification of Bacteria by Matrix-Assisted Laser Desorption Ionization Time-of-Flight Mass Spectrometry. *Clin. Infect. Dis.* **2009**, *49*, 543–551.

(9) Lagier, J.-C.; Edouard, S.; Pagnier, I.; others Current and Past Strategies for Bacterial Culture in Clinical Microbiology. *Clin. Microbiol. Rev.* **2015**, *28*, 208–236.

(10) van Veen, S. Q.; Claas, E. C. J.; Kuijper, E. J. High-Throughput Identification of Bacteria and Yeast by Matrix-Assisted Laser Desorption Ionization–Time of Flight Mass Spectrometry in Conventional Medical Microbiology Laboratories. *J. Clin. Microbiol.* **2010**, *48*, 900–907.

(11) Carbonnelle, E.; Mesquita, C.; Bille, E.; others MALDI-TOF Mass Spectrometry Tools for Bacterial Identification in Clinical Microbiology Laboratory. *Clin. Biochem.* **2011**, *44*, 104–109.

(12) Mather, C. A.; Rivera, S. F.; Butler-Wu, S. M. Comparison of the Bruker Biotyper and Vitek MS Matrix-Assisted Laser Desorption Ionization–Time of Flight Mass Spectrometry Systems for Identification of Mycobacteria Using Standardized Cultured Isolates. *J. Clin. Microbiol.* **2014**, *52*, 466–469.

(13) El Khéchine, A.; Henry, M.; Raoult, D.; Drancourt, M. Detection of Mycobacterium tuberculosis Complex Organisms in the Stools of Patients with Pulmonary Tuberculosis. *Microbiology* **2009**, *155*, 2384–2389.

(14) Berthet, F.-X.; Rasmussen, P. B.; Rosenkrands, I.; Andersen, P.; Gicquel, B. A Mycobacterium tuberculosis operon encoding ESAT-6 and a novel low-molecular-mass culture filtrate protein (CFP-10). *Microbiology* **1998**, *144*, 3195–3203.

(15) Sørensen, A. L.; Nagai, S.; Houen, G.; Andersen, P.; Andersen, A. B. Purification and Characterization of a Low-Molecular-Mass T-Cell Antigen Secreted by Mycobacterium tuberculosis. *Infect. Immun.* **1995**, *63*, 1710–1717.

(16) Mori, T.; Sakatani, M.; Yamagishi, F.; others Specific Detection of Tuberculosis Infection: An Interferon-γ-Based Assay Using New Antigens. *Am. J. Respir. Crit. Care Med.* **2004**, *170*, 59–64.

(17) Pai, M.; Denkinger, C. M.; Kik, S. V.; others Gamma Interferon Release Assays for

Detection of Mycobacterium tuberculosis Infection. *Clin. Microbiol. Rev.* **2014**, *27*, 3–20.

(18) Lotz, A. S.; Blackmore, T. M.; Dobson, R. C. J.; others Design of Experiments to Develop a Multi-Class Diagnostic Test for Fever of Unknown Origin Using Mass Spectrometric Detection of Serum Biomarker Panels. *J. Transl. Med.* **2010**, *8*, 117.

(19) Virtanen, P.; Gommers, R.; Oliphant, T. E.; others SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nat. Methods* **2020**, *17*, 261–272.

(20) Harris, C. R.; Millman, K. J.; van der Walt, S. J.; others Array Programming with NumPy. *Nature* **2020**, *585*, 357–362.

(21) Savitzky, A.; Golay, M. J. E. Smoothing and Differentiation of Data by Simplified Least Squares Procedures. *Anal. Chem.* **1964**, *36*, 1627–1639.

(22) Eilers, P. H. C.; Boelens, H. F. M. Baseline Correction with Asymmetric Least Squares Smoothing. *Leiden University Medical Centre Report* **2005**, Unpublished manuscript.

(23) Sauve, A.-C.; Speed, T. P. Normalization, Baseline Correction and Alignment of High-Throughput Mass Spectrometry Data. *Proc. 2004 CASE Workshop* **2004**, Technical report, UC Berkeley.

(24) Gross, J. H. *Mass Spectrometry: A Textbook*; Springer: Berlin, Germany, 2004.

(25) Broadhurst, D.; Goodacre, R.; Reinke, S. N.; others Guidelines and Considerations for the Use of System Suitability and Quality Control Samples in Mass Spectrometry Assays Applied in Untargeted Clinical Metabolomic Studies. *Metabolomics* **2018**, *14*, 72.

(26) Pedregosa, F.; Varoquaux, G.; Gramfort, A.; others Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.* **2011**, *12*, 2825–2830.

(27) Chen, T.; Guestrin, C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 2016; pp 785–794.

(28) Ke, G.; Meng, Q.; Finley, T.; others LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Adv. Neural Inf. Process. Syst.* **2017**, *30*, 3146–3154.

(29) Prokhorenkova, L.; Gusev, G.; Vorobev, A.; Dorogush, A. V.; Gulin, A. CatBoost: Unbiased Boosting with Categorical Features. *Adv. Neural Inf. Process. Syst.* **2018**, *31*.

(30) Brodersen, K. H.; Ong, C. S.; Stephan, K. E.; Buhmann, J. M. The Balanced Accuracy and Its Posterior Distribution. *Proc. Int. Conf. Pattern Recognit.* **2010**, 3121–3124.

(31) DeLong, E. R.; DeLong, D. M.; Clarke-Pearson, D. L. Comparing the Areas under Two or More Correlated Receiver Operating Characteristic Curves: A Nonparametric Approach. *Biometrics* **1988**, *44*, 837–845.

(32) Lundberg, S. M.; Lee, S.-I. A Unified Approach to Interpreting Model Predictions. *Adv. Neural Inf. Process. Syst.* **2017**, *30*.

Figure 1: Mean MALDI-TOF MS spectra (±1 SD) for TB (blue) and NTM (orange) isolates after preprocessing, over the 3–15 kDa range. Insets show expanded views of the (a) CFP-10/CFP-10* region (9.8–11.0 kDa) and (b) ESAT-6/ESAT-6* region (7.7–10.0 kDa). Dashed vertical lines indicate the theoretical m/z of all 12 targeted biomarker features (Table 2).

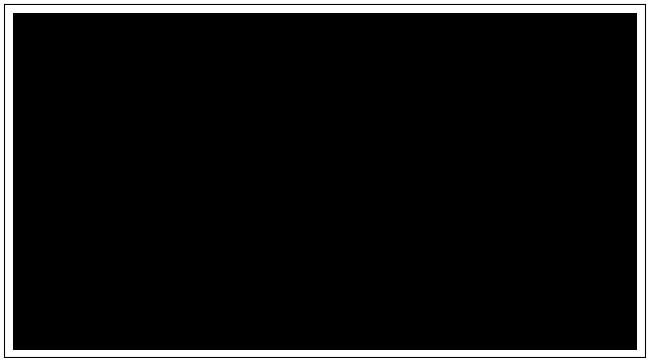
Figure 2: Sequential preprocessing of a representative TB MALDI-TOF spectrum. (a) Raw spectrum; (b) after Savitzky–Golay smoothing (window = 21, order = 3); (c) CLSA baseline estimate (red) overlaid on smoothed spectrum; (d) baseline-subtracted and TIC-normalized spectrum with biomarker extraction windows highlighted. The $\sqrt{m/z}$ transformation applied within the CLSA algorithm normalizes peak widths across the mass range, preventing under-correction at higher masses.

Figure 3: Distribution of the 12 biomarker feature intensities (after TIC normalization and local background subtraction) for TB (blue) and NTM (orange) classes. Boxes indicate median and interquartile range; whiskers extend to $1.5 \times$ IQR. Asterisks indicate statistical significance after Bonferroni correction ($^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; Mann–Whitney U test). Fold changes (TB/NTM mean) are reported above each panel.

Figure 4: Principal component analysis (PCA) of the 12-biomarker feature matrix. Each point represents one spectrum, colored by diagnostic class (TB: blue; NTM: orange). Ellipses represent the 95% confidence region assuming bivariate normality. Percentages indicate variance explained by PC1 and PC2.

Figure 5: Cross-validation AUROC for all 43 machine learning configurations, grouped by algorithm family. Points indicate the mean of the 5 folds; error bars indicate ± 1 SD. Colors distinguish algorithm families as indicated in the legend. The dashed horizontal line marks the performance of the stratified dummy classifier (no-information baseline).

Figure 6: Independent test set performance of the three top-ranked classifiers. (a–c) Confusion matrices (rows: true label; columns: predicted label); values are counts and row-normalized percentages. (d) Receiver operating characteristic (ROC) curves with 95% confidence bands computed by DeLong bootstrapping. Diagonal dashed line: no-discrimination reference.

Figure 7: SHAP beeswarm plot for the CatBoost classifier on the independent test set. Each point represents one spectrum; the x-axis shows the SHAP value (contribution to the log-odds of a TB prediction); color encodes the feature value (blue: low; red: high). Features are ordered by mean absolute SHAP value (top = most important).

TOC Graphic